

TRANSPORT PHENOMENA II, CHE 312

QUIZ 1

CLOSED BOOK, TIME: 15 MINUTES

A liquid with a plug flow velocity profile $v_z(r) = v_0$ is being heated within a cylindrical tube of radius R with a constant heat flux $q_r(z, r=R) = q_R < 0$.

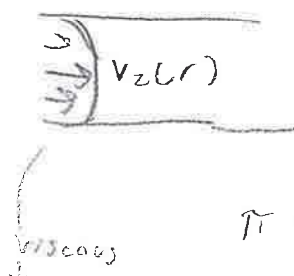
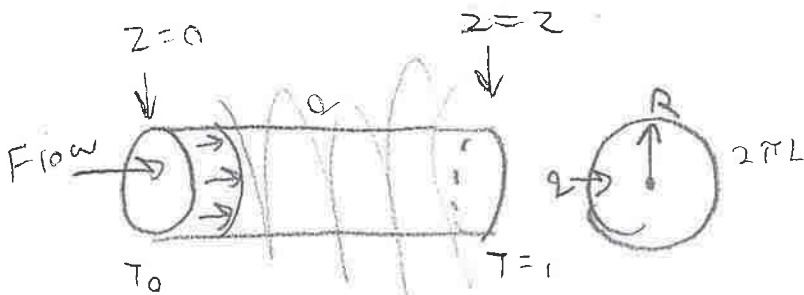
↳ Heating

Volume:

Model assumption:

- Steady-State
- Constant physical properties ρ, c_p, k .
- $c_v \approx c_p$ is constant and $\frac{du}{dT} = c_p$
- Heat flux in the axial direction is negligible. $q_z = 0$ $q_r =$
- Negligible viscous heating.
- Fluid enters the heating region of the tube with a uniform temperature $T(z=0, r) = T_1$

Derive an expression for the bulk (cup-mixing average) temperature of the fluid as a function of z .



$$\pi = \tau + \rho \partial_{ij}$$

$$\tau_{rz} = -\mu \left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z} \right)$$

$$\tau = -\mu \left(\frac{\partial v_z}{\partial r} \right)$$

$$0 = -\int dS \langle q_{in} \rangle + -\int dS \langle v \cdot \pi \rangle$$

$$\int_0^{2\pi} \int_0^R r q_r dr$$

$$\int_0^{2\pi} \int_0^L z$$

$$2\pi r^2 q_r|_{r=0} = 2\pi r^2 q_r|_{r=R} + \mu \frac{\partial v_z}{\partial r}$$

$$-2\pi r^2 k \frac{\partial T_0}{\partial r} = -2\pi r^2 k \frac{\partial T_1}{\partial r}$$

interpretation wrong

r - ok

z - missing

maths looking

T_0